

Scalable Data Engineering Solution

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Uppsala University

Department of Information Technology

Team Members: Group 32

Abdur Rehman Khalid	abdur-rehman.khalid.8716@student.uu.se
Arnab Kumar Ghosh	arnab-kumar.ghosh.4724@student.uu.se
Dip Chowdhury	dip.chowdhury@student.uu.se
Muhammad Umair	muhammad.umair.6023@student.uu.se
Prodip Kumar Das	prodip-kumar.das.8559@student.uu.se

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1 Background

Urban transportation systems generate vast volumes of data that can be leveraged to improve traffic management, infrastructure planning, and mobility services. One of the most widely used real-world datasets in this domain is the New York City Taxi Trip dataset, published by the New York City Taxi and Limousine Commission (TLC) [1]. This dataset contains detailed records of taxi trips, including pickup and drop-off timestamps, geographic zones, trip distances, and fare amounts, making it a valuable resource for studying large-scale urban mobility patterns.

The NYC taxi dataset is particularly relevant for data engineering research due to its scale and complexity. In this project, a subset of approximately 10–20 GB of data stored in Apache Parquet format is used. Columnar storage formats such as Parquet are designed for efficient analytical workloads by enabling better compression and reducing I/O during query execution, which is critical for large-scale data processing systems. The dataset contains millions of trip records, making it suitable for evaluating distributed systems under realistic conditions. Such large-scale data processing challenges are commonly addressed using parallel computing paradigms like MapReduce [2], which form the foundation of modern distributed frameworks.

Prior research has demonstrated the usefulness of taxi trip data in applications such as demand prediction, traffic flow analysis, and anomaly detection. For instance, studies in urban computing have used mobility datasets to identify spatial and temporal demand patterns, enabling improved resource allocation in cities [3]. Other works have explored predictive modeling techniques to estimate trip duration and detect irregular travel behavior. These examples highlight that beyond analytical insights, such datasets are also valuable benchmarks for evaluating the performance of large-scale data processing systems.

In this project, the analytical objective is intentionally simple: to compute the average trip duration and average fare amount grouped by pickup zone and hour of the day. This type of aggregation represents a common workload in data engineering pipelines and involves filtering, grouping, and aggregation operations over large datasets. Keeping the analytical task simple ensures that the focus remains on system performance and scalability rather than algorithmic complexity, which aligns with the project's emphasis on engineering aspects.

To efficiently process the dataset, the project utilizes Apache Spark, a distributed data processing engine designed for large-scale analytics [4], in combination with Hadoop Distributed File System (HDFS) for scalable and fault-tolerant storage. These technologies enable parallel execution across multiple nodes, allowing the workload to be distributed and processed efficiently. The system is deployed on a constrained cluster environment consisting of virtual machines, reflecting realistic limitations in computational resources and reinforcing the importance of efficient system design.

A central focus of this project is the evaluation of system scalability through both horizontal and vertical scaling experiments. By varying the number of worker nodes and computational resources, the project analyzes performance improvements and identifies bottlenecks such as network overhead, data shuffling, and resource contention. The observed results are interpreted using theoretical models such as Amdahl's Law [5], which provides a foundation for understanding the limits of parallel speedup. This approach ensures that the project not only demonstrates practical system implementation but also critically evaluates the behavior of distributed systems under realistic big data workloads.

2 Data Format & Characteristics

[Describe the dataset technically: format (CSV, JSON, Parquet, etc.), schema/structure, total size and number of records, data quality issues encountered, and justification for format choice or conversion.]

3 System Architecture & Implementation

This section details the system architecture, infrastructure deployment, and technical implementation of our robust data pipeline. We outline the hardware topology deployed on OpenStack, why we selected our core software stack, and the engineering mechanisms driving our ETL and analysis jobs.

3.1 System Design

Our system follows a classic centralized-master, distributed-worker model tailored for batch processing big data. As depicted in Figure 1, the architecture is divided into three primary layers: Ingestion, Distributed Storage (HDFS), and Distributed Processing (Apache Spark). Data is fetched externally into the master node, pushed to HDFS, and subsequently processed by Spark executors dispersed across the worker nodes.

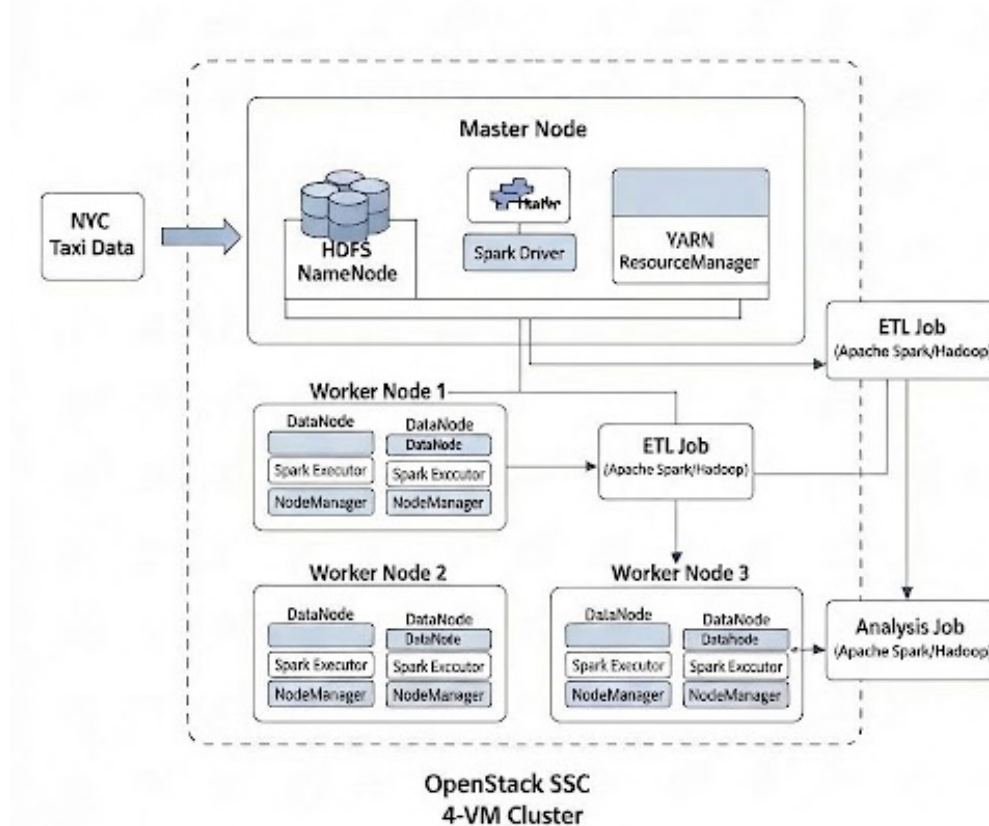


Figure 1: System architecture diagram showing the OpenStack deployment, node topology, and the internal data pipeline flow from raw Parquet to aggregated results.

3.2 Technology Stack

Given the project requirements, we adopted a modern big data stack focused on in-memory processing capabilities:

- **Storage (HDFS via Hadoop 3.4.1):** Chosen for its fault tolerance and native integration with Spark. While cloud object storage (like AWS S3) is popular today, deploying HDFS on bare VMs offers direct control over data locality and block replication factors.
- **Processing (Apache Spark & PySpark 3.5.1):** We selected Spark over standard Hadoop MapReduce because Spark holds intermediate RDDs and DataFrames in mem-

ory, drastically accelerating the iterative transformations required during our ETL phase. PySpark provides an accessible DataFrame API and straightforward type-casting structures.

- **Orchestration & Resource Management:** We relied on standalone cluster management managed heavily by Bash scripts (`run_benchmark.sh` and `deploy.sh`) combined with Git for CI/CD, creating a lightweight yet effective pipeline orchestration.

3.3 Infrastructure Configuration

Our cluster is deployed entirely on the Swedish National Infrastructure for Computing (SNIC) OpenStack instance. The infrastructure consists of 4 tightly coupled Virtual Machines (VMs):

- **Master Node (1x ssc.medium):** 2 vCPU, 4 GB RAM. This node is assigned a Floating IP for external SSH access. It houses the Hadoop NameNode, ResourceManager, and the Spark Driver process.
- **Worker Nodes (3x ssc.medium):** 2 vCPU, 4 GB RAM each. These sit on a private sub-network (192.168.2.x subnet). They run the Hadoop DataNodes, NodeManagers, and Spark Executor instances.

In total, the cluster harnesses 8 vCPUs and 16 GB of RAM, capped with a strict 100 GB collective storage limit. To maintain data reliability without exceeding storage quotas, HDFS was configured with a replica factor of 2 (`dfs.replication=2`).

3.4 Data Pipeline

The pipeline executes in three synchronized stages:

1. **Ingestion:** Raw NYC Taxi Parquet files are downloaded from the TLC portal to the Master node via `wget` and deposited straight into `hdfs://.../data/nyc-taxi/`.
2. **ETL Phase:** The `etl_job.py` script ingests the raw data block by block. It filters out invalid records (e.g., zero-distance trips, null metrics), converts timestamps to calculate `duration_minutes` and `pickup_hour`, aggressively prunes dropped columns, and commits the clean records back to HDFS at `/data/nyc-taxi-cleaned/`.
3. **Analysis Phase:** Finally, `analysis_job.py` aggregates the validated dataset. It groups by `PULocationID` and `pickup_hour` to compute the average fare and trip duration. It subsequently performs a left join against a static CSV (`taxi_zone_lookup.csv`) to convert abstract ID numbers into human-readable zone names, coalescing the final output into a single file for stakeholder ingestion.

3.5 Code Structure

The codebase follows a modular design paradigm:

- **src/config.py:** Centralizes cluster endpoints, absolute HDFS URI paths, and basic threshold configurations to eliminate hardcoded parameters across jobs.
- **src/etl_job.py:** The heaviest processor. Given resource constraints, it bypasses standard `spark.read.parquet("*.parquet")` schema inferences and iteratively targets singular files using `subprocess.run` to interact with HDFS.
- **src/analysis_job.py:** Houses the PySpark SQL aggregations, strictly outputting to a targeted CSV path using `coalesce(1)`.

- **scripts/run_benchmark.sh**: A specialized harness that automates execution over varied core allocations (`-total-executor-cores`) and node arrays to capture timestamps for plotting horizontal and vertical scales.

3.6 Challenges & Solutions

We encountered several prominent technical challenges during development:

1. **Dataset Schema Volatility**: When reading multi-month TLC data, PySpark's vectorized Parquet reader routinely crashed (`[CANNOT_MERGE_SCHEMAS]`) due to invisible shifts in the column type for `Airport_fee` (varying from `INT32` to `FLOAT64` dynamically by month). *Solution*: We restructured the ETL job into a `for`-loop that asks HDFS for the explicit list of file paths. Spark reads them independently, safely infers local schemas, selectively prunes the offensive column, and appends into a normalized structure.
2. **Strict Storage Exhaustion**: 25 GB of storage per VM meant HDFS was easily overwhelmed caching uncompressed RDDs and saving multiple layers of Parquet processing data. *Solution*: We established an aggressive "select and drop" mechanism in the ETL. We discard roughly 15 unused columns representing several gigabytes of overhead, saving only the 4 tightly packed elements (`PULocationID`, `fare_amount`, `pickup_hour`, `duration_minutes`) needed for analysis.
3. **Dynamic Allocation Skewing Benchmark Constraints**: Spark's dynamic resource allocation skewed the early iterations of our benchmarking by automatically distributing workloads across nodes we were attempting to restrict. *Solution*: We disabled dynamic allocation directly via PySpark config keys (`spark.dynamicAllocation.enabled=false`) and enforced hard caps using Spark submit metrics.

4 Scalability Experiments & Results

[Present experimental findings: experimental design, results with plots, analysis of results, bottleneck identification, and comparison to theoretical expectations.]

4.1 Experimental Design

[Describe what you varied in experiments (number of workers, cores, data size), the metrics measured (runtime, throughput), and the methodology used.]

4.2 Horizontal Scaling Results

[Present results of horizontal (strong) scaling experiments: fixed dataset size with increasing number of worker nodes. Include runtime vs. number of workers plot.]

4.3 Vertical Scaling Results

[Present results of vertical scaling experiments: fixed number of nodes with increasing cores/memory per node. Include runtime vs. resources plot.]

4.4 Speedup & Efficiency Analysis

[Compute and plot the speedup factor ($S = T_1/T_N$) and efficiency ($E = S/N$). Compare against ideal linear speedup. Discuss deviations.]

4.5 Bottleneck Identification

[Identify what limits scalability: CPU, network I/O, disk I/O, shuffle overhead, data skew, or serialization. Reference Amdahl's Law if applicable.]

5 Discussion & Conclusion

[Reflect on the project: Was the architecture suitable? What worked well? What would you improve with more time/resources? What did you learn about distributed systems?]

References

- [1] New York City Taxi and Limousine Commission, *New York City Taxi and Limousine Commission (TLC) Trip Record Data*, Accessed: 2026-03-01, 2024. [Online]. Available: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
- [2] J. Dean and S. Ghemawat, “Mapreduce: Simplified data processing on large clusters,” *Communications of the ACM*, vol. 51, no. 1, pp. 107–113, 2008.
- [3] G. Andrienko, N. Andrienko, P. Bak, D. Keim, and S. Wrobel, “Visual analytics of movement,” 2013.
- [4] M. Zaharia, M. Chowdhury, M. J. Franklin, S. Shenker, and I. Stoica, “Apache spark: A unified engine for big data processing,” *Communications of the ACM*, vol. 59, no. 11, pp. 56–65, 2016.
- [5] G. M. Amdahl, “Validity of the single processor approach to achieving large scale computing capabilities,” *Proceedings of the April 18-20, 1967, spring joint computer conference*, pp. 483–485, 1967.